

HD-grade movies and life-like games will deliver nirvana to their audiences, but they're resource-hungry; memory-hungry in particular

Intel is in a strong enough position to dictate terms to memory manufacturers. Other giants including Microsoft are also in favour of this move; we already know Vista is a memory bandwidth hog.

Then there is NRAM or Nano RAM, which promises much higher densities. NRAM is non volatile unlike conventional RAM, and is based on the positions of carbon nanotubes on a chip-like substrate. However, the development is largely in conceptual stages. The benefits would be higher memory densities (and therefore cheaper) and low power consumption. Another advantage than NRAM has over Flash is the much longer life cycle of the insulators used.

Yet another technology that shows promise is MRAM (Magnetoresistive RAM) which works on the principle of magnet-induced storage rather than electric charges or current flows like in current DRAM. It's non-volatile, but doesn't degrade like Flash does. Speeds are right up there with latencies of 2ns already demonstrated. MRAM is already present in cer-

tain very niche markets, though the initial pricing has been a deterrent.

Graphics memory has got a further shot in the arm with better GDDR3 yields, allowing for memory speeds of up to 2 GHz. The major players in the GDDR3 market are Infineon, Samsung, Hynix, and Micron. Samsung is pushing their GDDR4 memory, which we've already seen in action on the ATI X1950XTX. The main benefit of GDDR4 is reduced power consumption, a smaller packaging, and, of course, greater bandwidth and higher speeds (up to 2.6 GHz).

The future of multimedia is HD-grade movies and life-like games. These applications will deliver nirvana to their audiences, but they're resource-hungry; memory-hungry in particular. Gamers in the States have already moved to high-speed, 4 GB, dual-channel solutions. In India we've resisted the urge to splurge—restrictive prices and insufficient knowledge about the benefits of more memory being the culprits. Let's hope we can play a part in changing things starting from here. ☒

Storage

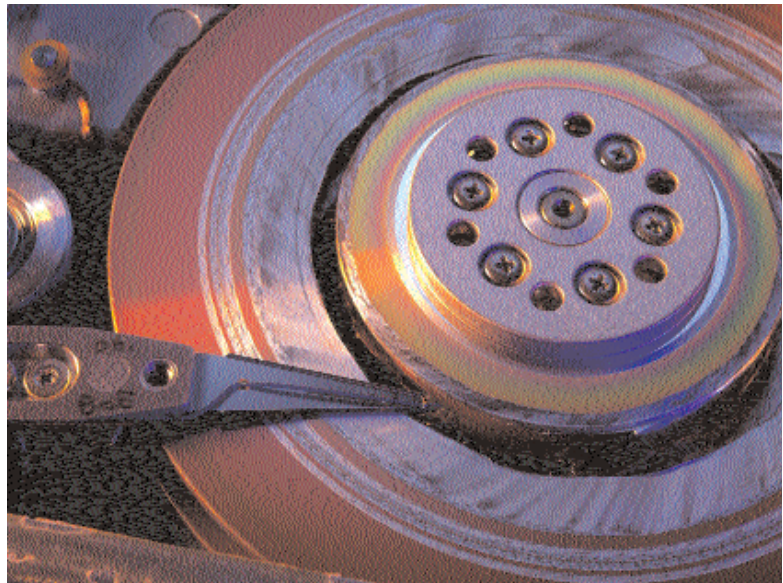
Windows Vista will demand more from your hard disks—not to mention all those HD movies and music you're downloading

History proves our species is one of incurable hoarders... from precious metals to food, from fuel to weapons. There is something else mankind has hoarded through the ages—information. The digital era has pandered to this whim, even encouraged it.

Steady advancements in drive technology have seen sharply declining prices is another significant factor. A 250 GB Desktop hard drive retails at around Rs 3,500 today, while a 160 GB drive costing in excess of Rs 4,500 a couple of years ago is available for as little as Rs 2,700 now. Price per GB is down to rupees 16.875 per GB or 0.01 paisa per MB! Larger drives have also become cheaper, a 400 GB drive costs under 5,000 rupees now, while the rupees 10,000 plus costing 500 GB drives are near the 7,000-rupee mark.

Serial ATA (SATA) has matured as an interface, quickly becoming de facto for Desktop hard drives. Perpendicular Recording technology has emerged, leading to even greater capacities due to increased areal density. The 750GB barrier was hit last year (Seagate), and this year Hitachi has hit 1 Terabyte. The 1 Terabyte Hitachi features a 32-megabyte cache, and five platters—each capable of 200 GB. Though the initial cost of cramming so much data per platter is high, acceptance in the market will cause economies of large-scale production to take over.

A new hard drive technology called Hybrid Hard Drive or HHD has emerged, featuring a



very high capacity (1 GB), high-speed, Flash-based buffer. The colossal cache means less spinning up and down of platters—power saving and higher performance. In the normal state, the platters are at rest and all data is written to the cache. The platters only spin when cache has run out, or when data needs to be read from the platters. Coupled with the fallen flash prices we could well see hard drives with 8 and 16 GB caches.